Taiki Aiba

taiba3@gatech.edu — 978-201-4007 — linkedin.com/in/taiba3 — detoasty3.github.io

EDUCATION

Georgia Institute of Technology, Atlanta, GA – GPA: 3.92/4.0 — May 2027
M.S. in Mathematics, M.S. in Computer Science

Georgia Institute of Technology, Atlanta, GA – GPA: 3.96/4.0 — August 2025
B.S. in Mathematics, B.S. in Computer Science, B.S. in Applied Languages and Intercultural Studies, Minor in Economics
Studied abroad at **Ritsumeikan Asia Pacific University**, Beppu, Oita, Japan in Summer 2025

RESEARCH EXPERIENCE

Polymath Jr REU, **Williams College**, Remote — Summer 2024, 2025, 2026
Advisors: Timothy Goldberg (Lenoir-Rhyne University), Lauren Rose (Bard College), Hema Gopalakrishnan (Sacred Heart University)

- Summer 2024: Researched the quad number $q(n)$ in the card game *Quads*. Studied affine geometry and *Quads* cards as bit vectors in $\mathbb{Z}_2^6$. Wrote C++ code to brute-force results for $n \in [4, 16]$ and $d = \lceil \log_2 n \rceil$; used these results to conjecture that d does not matter and $q(2^n + 1) = q(2^n)$ for all n and to prove an upper bound expression for $q(n)$ and attainability for n a power of two.
  - Wrote the unpublished paper *Quad Packing* (paper). Presented at JMM 2025 and MAA-SE 2025 (slides).
- Summer 2025, 2026: Mentored 20 undergraduates, introducing them to the research problem and the game itself through resources and weekly meetings. Proved an exact expression for the flat (r-dimensional affine subspaces of a vector space V) number, showing that each term $M(n)$ can be written as the sums and products of certain combinations of powers of 2 in the binary representation of n. Proved quad packing is equivalent to flat packing via an injective mapping, resolving a conjecture in a 2023 MIT PRIMES paper.
  - Resulted in *Quad-Packing in the Game EvenQuads* (WIP). Presented at MAA-SE 2026 (slides).

Georgia Tech School of Mathematics, Atlanta, GA — Fall 2023 - Spring 2026
Advisor: Ernie Croot

- Fall 2023 - Spring 2024: Explored the Prouhet-Tarry-Escott problem and attempted to find results using contour integration and algebraic bounding. Explored the Kirchhoff Matrix-Tree Theorem and researched domino tiling enumerations on a grid to attempt to strengthen existing bounds. Studied grid-counting problems (e.g., rook problem) and searched for bijections to spanning trees (á la Temperley's bijection) in order to apply the Kirchhoff Matrix-Tree Theorem (used this problem to transition to researching grid graphs).
- Fall 2024 - Spring 2025: Researched maximal induced forests in multi-dimensional grid graphs and found and proved upper and lower bounds on d dimensions with the hand-shaking lemma and considering sums of coordinates mod $2d + 1$. Found a zig-zag construction giving a tight upper bound of $(2n^2 + O(n))/3$ for two-dimensional graphs.
  - Wrote the unpublished paper *Maximal induced forests in grid graphs*, but scrapped it after discovering our results were already found in the paper *New bounds on the size of the minimum feedback vertex set in meshes and butterflies* by Caragiannis *et al.* (2002).
- Summer 2025 - Spring 2026: Found an upper bound of $3n^2/4$ for the size of largest induced 4-cycle-free subgraphs in two-dimensional grid graphs by taking 2×2 subgrids and orienting them the same way. Found an upper bound of $(n/2)^n$ for the number of largest induced 4-cycle-free subgraphs in two-dimensional grid graphs by considering local 2×2 subgrids and identifying a necessary constructive condition. Proved upper and lower bounds on the number of such subsets with $(3/4 - \varepsilon)n^2$ vertices with respect to ε by counting based on local $2k \times 2k$ subgrids and upper bounding binomial coefficients with Stirling bounds.
  - Resulted in *4-cycle-free induced subgraphs of grid graphs* (arXiv, submitted to *Ars Combinatoria*). Presented at MAA-SE 2026 (slides).

Georgia Tech School of Modern Languages, Atlanta, GA — Summer 2024
Advisor: Kyoko Masuda

- Researched frequency of female versus male pronouns in the manga *Sazae-san* and drew conclusions based on different age groups.
  - Submitted collaborative research paper in Japanese to the SGSJ in 2024. Presented at SGSJ 2024.

Georgia Tech School of Electrical and Computer Engineering, Atlanta, GA — Fall 2023
Advisor: Aaron Lanterman

- Translated documents of the PC-FX game console from Japanese to English, compiled in a GitHub repository for general usability. Analyzed the C Compiler of the GMAKER Starter Kit (processing flow/registers), allowing user-made software to run.

READING EXPERIENCE

Georgia Tech School of Mathematics, Atlanta, GA — Spring 2025
Advisor: Xingxing Yu

- Wrote the unpublished paper *On the Turán Number of the Blow-Up of the Hexagon* (paper), surveying Janzer *et al.* (2022). Gave a one-hour whiteboard presentation to a class of ∼10 students; collaborated with a Ph.D. student and a post-doctorate.

Independent Reading, Atlanta, GA — Summer 2023

- Read about Burnside's lemma, including applications to classic problems and competition problems (AMC/AIME).
  - Wrote the unpublished paper *Burnside's lemma and applications in competition problems* (paper). Presented at MAA-SE 2026 (slides).

Georgia Tech Directed Reading Program, Atlanta, GA — Fall 2022 - Spring 2025

- Read and presented on generating functions in combinatorics and algebra in Fall 2022, combinatorial problem solving techniques in Spring 2023 (slides), the Union-Closed Sets Conjecture in Fall 2023 (slides), Sudoku and graph theory in Spring 2024 (slides), quantitative finance in Fall 2024, and data structures/algorithms, focusing on the Gomory-Hu Tree, in Spring 2025 (slides).

AWARDS

Polymath Jr REU Mentor Stipend ($2500) May 2026
Awarded to one graduate mentor per project. In Summer 2026, I was the paid mentor for the Games and Finite Geometry project.

Outstanding Graduating Mathematics Major ($100) April 2025
Awarded to select outstanding undergraduate senior mathematics majors annually. In 2025, I was one of five award recipients.

Outstanding Senior in Japanese April 2025
Awarded to one outstanding undergraduate senior annually in the Japanese department.

University of Illinois Freshman Math Contest Third Place October 2021
Ranked third out of 22 first-years in a five-question proof-based math contest across the three University of Illinois campuses.

TEACHING

Georgia Institute of Technology, Atlanta, GA

- CS 6515: Introduction to Graduate Algorithms (Teaching Assistant: Spring 2027)
- CS 3510: Design and Analysis of Algorithms (Teaching Assistant: Fall 2024, Spring 2025, Fall 2025, Spring 2026)
- CS 2050: Introduction to Discrete Mathematics for Computer Science (Teaching Assistant: Summer 2023, Fall 2023, Spring 2024)
- MATH 2551: Multivariable Calculus (Lecture Assistant: Spring 2023)

University of Illinois Urbana-Champaign, Urbana, IL

- CS 124: Introduction to Computer Science I (Course Assistant: Spring 2022)

WORK EXPERIENCE

Capital One, Software Engineer, Location TBD Starting August 2027

- Incoming full-time software engineer.

IBM, Software Engineer Intern, San Jose, CA August 2026 - December 2026

- Incoming intern in IBM watsonx.governance.

Capital One, Software Engineer Intern, McLean, VA June 2026 - August 2026

- Automated data usage validation in card acquisitions using Golang, slashing lookup time in QA by 98% by replacing manual log parsing.

Amazon, Software Engineer Intern, Boston, MA June 2024 - August 2024

- Developed a discrete event simulation model for packet processing in Alexa's nodes using C++ and SimPy to visualize latency trends.

Art of Problem Solving, Math Teaching Assistant, Waltham, MA July 2022 - Present

- Assist in weekly office hours, helping 200+ math students with extracurricular math. Assisted in 2022 and 2025 summer camps in-person.

Math League, LLC, Curriculum Specialist, Remote July 2021 - Present

- Write and edit problems for contests taken by 1000+ students worldwide. Taught the 2022-23 AfterMath classes, led the NS Challenge.

SERVICE

Mathematical Association of America, AMC 10/12 Editorial Board Member, Remote September 2024 - Present

- Propose problems for the AMC 10/12 and attend meetings to discuss proposals and determine which problems are suitable.

MATHCOUNTS Foundation, Question Writing Committee Member, Remote February 2025 - Present

- Propose problems for the MATHCOUNTS Competition Series and attend in-person meetings to review and edit the problems. Volunteer at the annual MATHCOUNTS National competition held in-person.

Mustang Math, Problem Writing Team Member, Remote July 2024 - Present

- Assist with hosting math contests primarily aimed at middle school students. Propose, edit, and review problems for the contests.

De Mathematics Competitions, Founder, Director, and Tutor, Remote September 2020 - Present

- Write and direct mock American Mathematics Competitions contests on the website Art of Problem Solving. Tutor a third-grade student for the AMC 8 since April 2025, a contest primarily aimed at middle school students.

Georgia Tech Undergraduate Mathematics Advisory Committee, President, Atlanta, GA Fall 2023 - Spring 2025

- Hosted 2-4 social events (e.g., puzzle tournaments, graduate student dinners, parties) for math majors every semester. Led and handled event logistics, redesigned and edited the UMAC website with photos, and designed flyers for all events.

Georgia Tech High School Math Day, Student Volunteer, Atlanta, GA Spring 2023, 2024, 2025, 2026, 2027

- Wrote problems for and volunteered (proctor, collect papers, grade) in-person at a contest taken by 300+ high school students annually.

Georgia Tech Japan Student Association, Executive Board Officer, Atlanta, GA Fall 2023 - Spring 2027

- Led weekly conversation practices to enrich Japanese learners' conversation skills. Hosted events related to the Japanese culture.

SKILLS

Programming: C/C++, Python 3, Java, Go, Rust, JavaScript/TypeScript, SQL, R, Verilog/SystemVerilog
Machine Learning: NumPy, pandas, PyTorch, TensorFlow, scikit-learn, SimPy, Matplotlib
Languages: English (native), Japanese (advanced), Spanish (beginner)
Interests: Set (winner of JMM 2025 Set Championships), drawing, blogging, electric guitar (9 years), piano (9 years), competitive programming